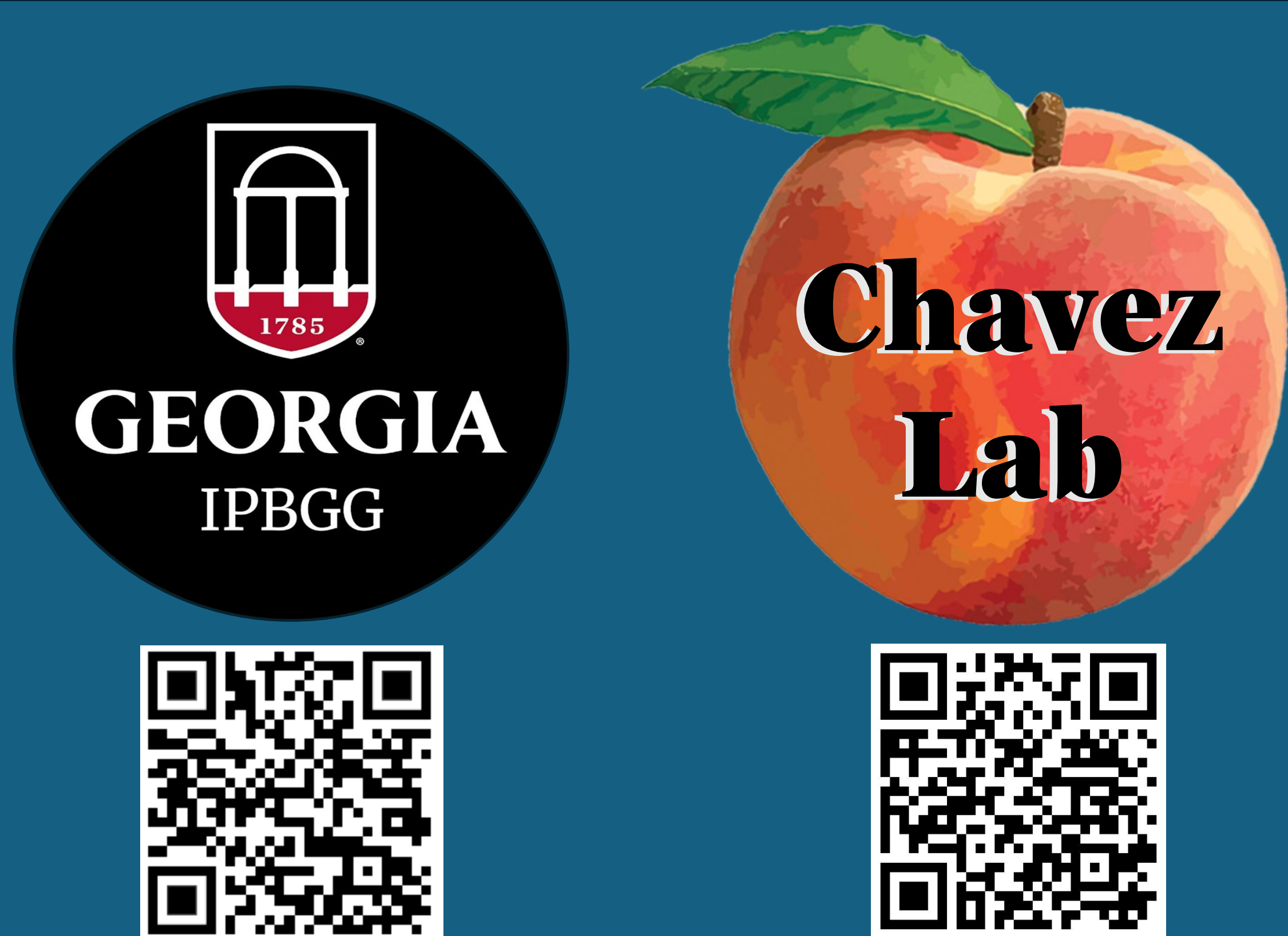




Capture-Seq technology as a genotyping and SNP discovery tool in peach

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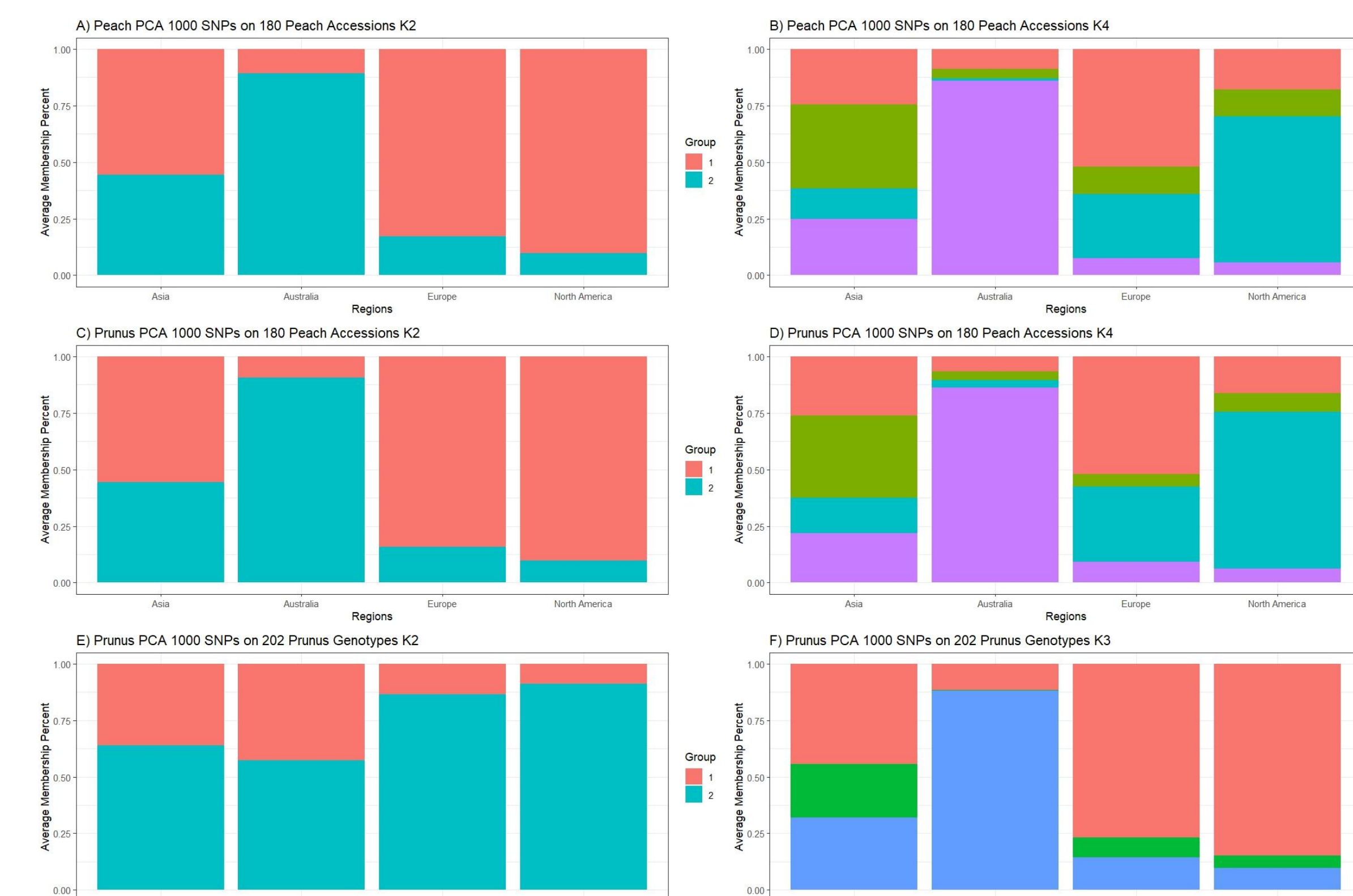
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Personal Website



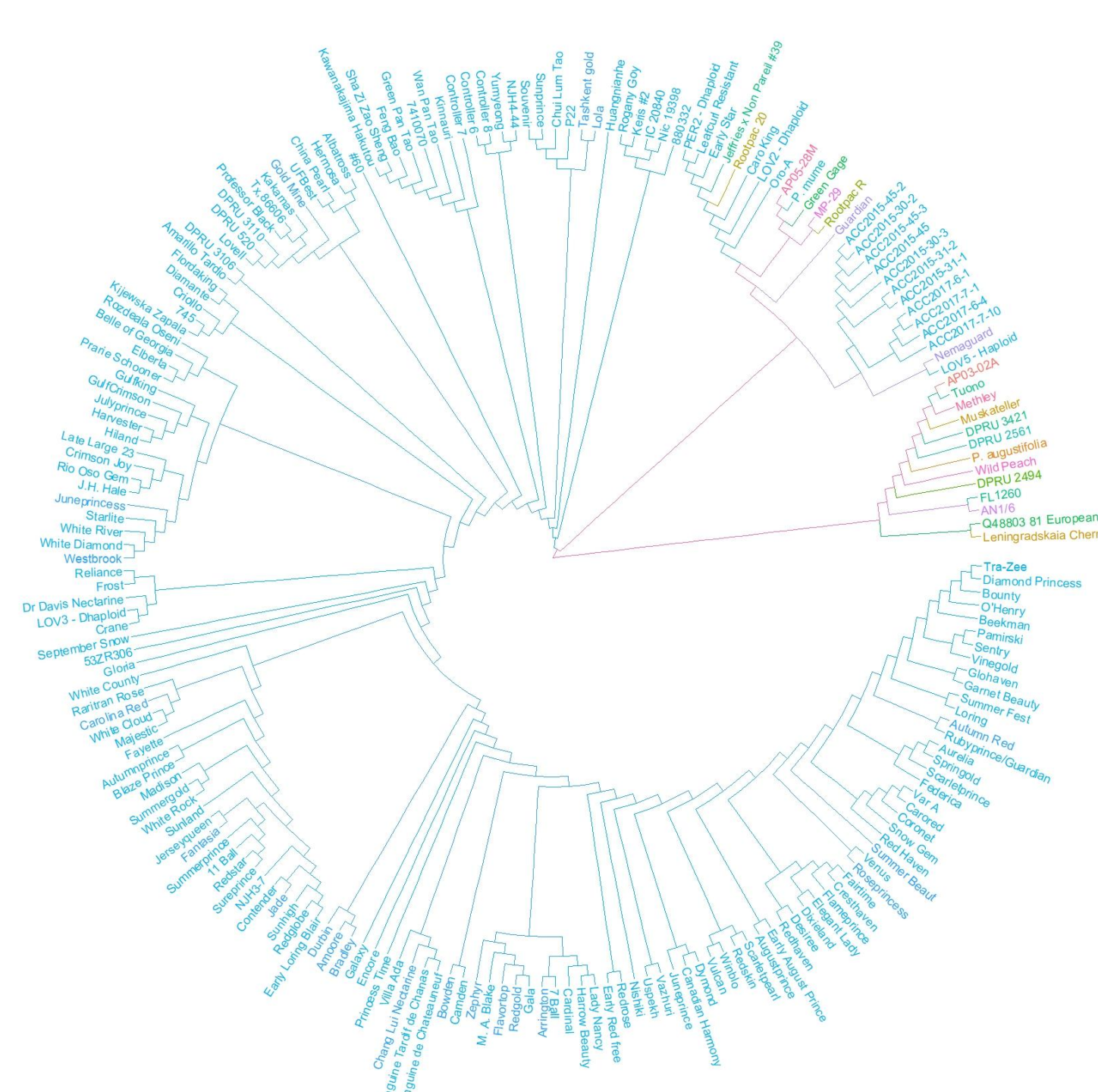
STRUCTURE Analysis – Group Membership By Population for Different K Values



UPGMA Phylogenetic Tree - Regions



UPGMA Phylogenetic Tree - Species



Background

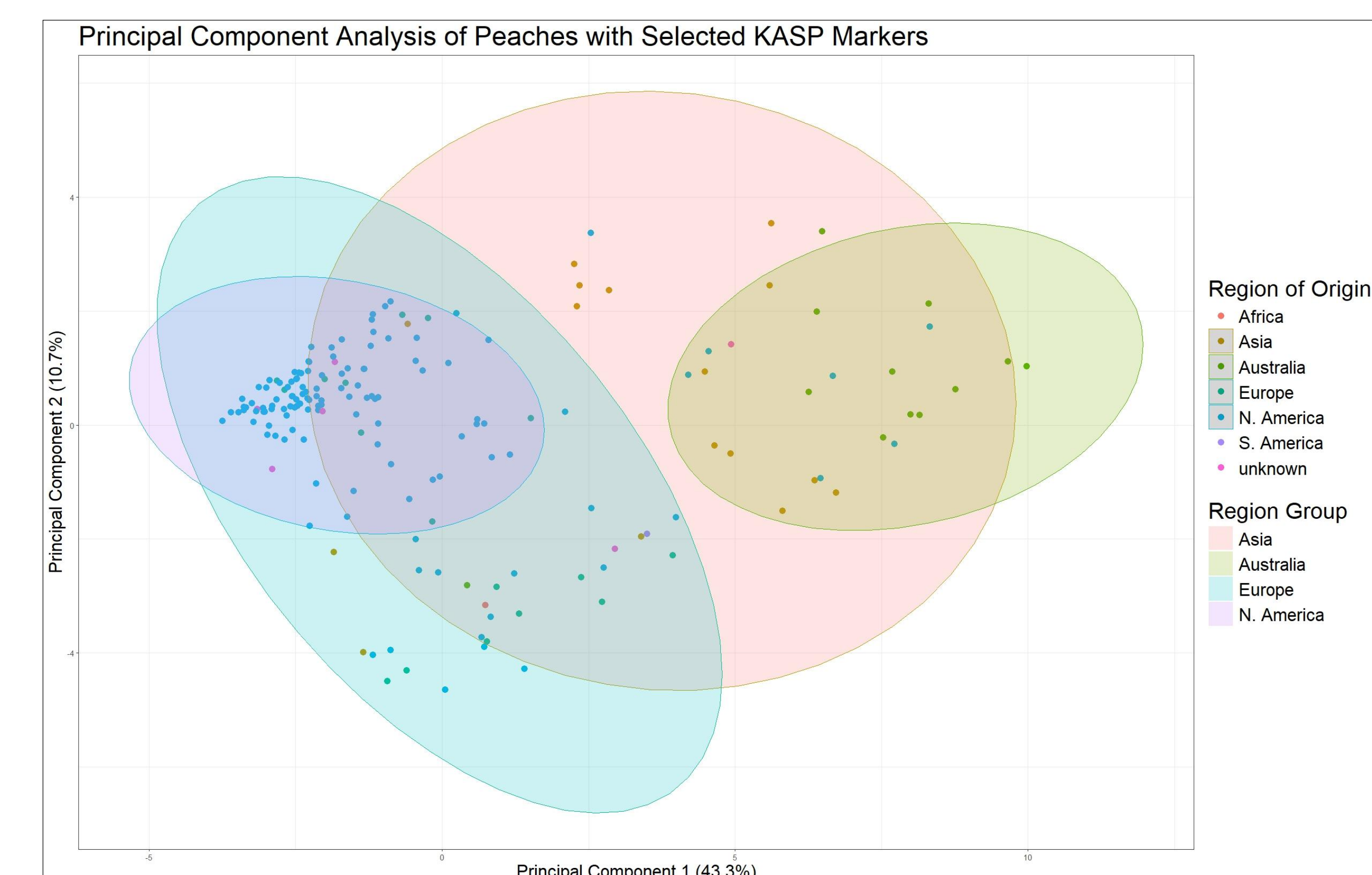
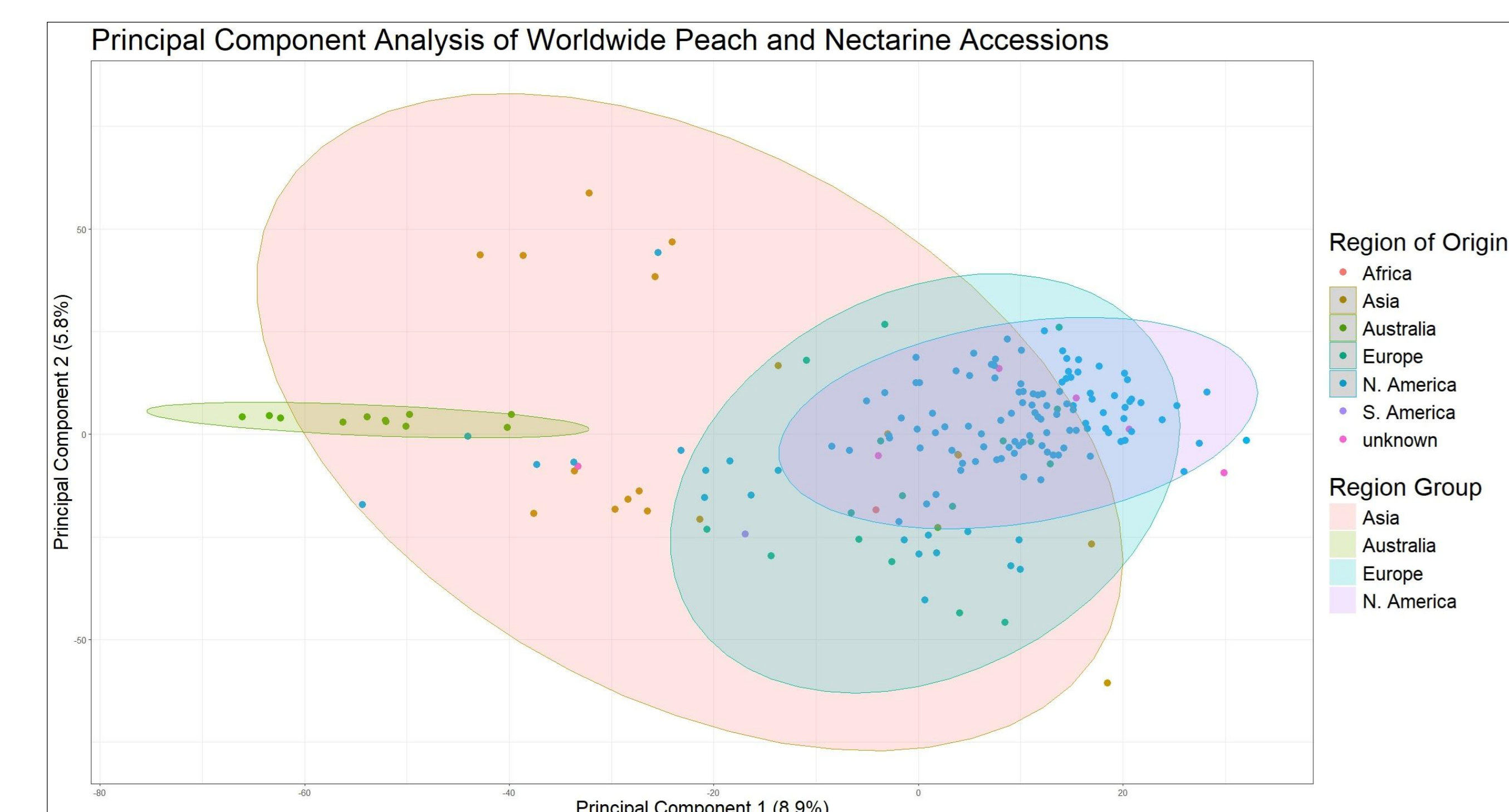
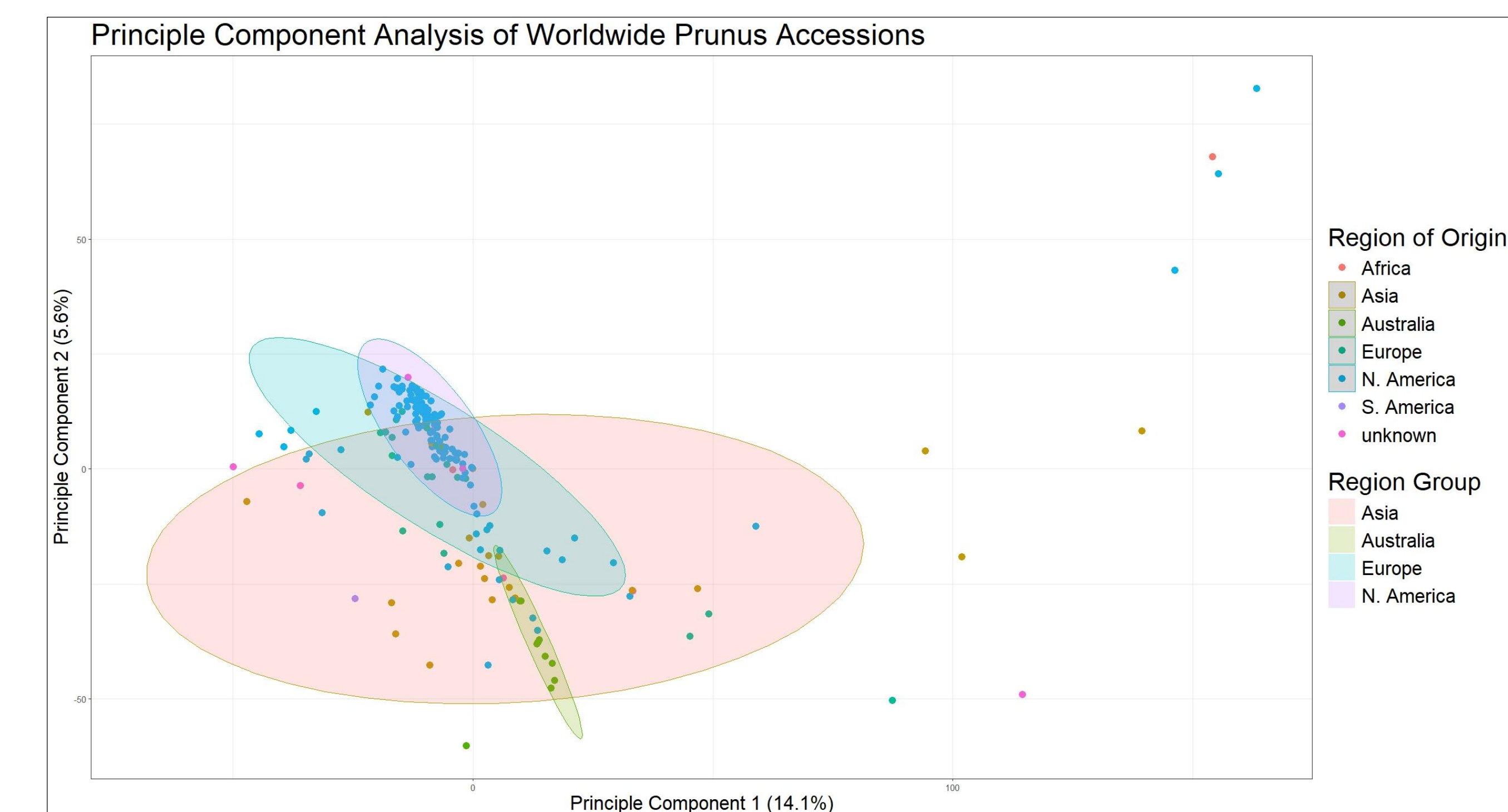
- Peaches in the US have a narrow genetic base.
 - A genetic bottleneck occurred in the 1870's, leaving peach production in the US vulnerable.
- Introducing new genetic material could bolster peach production.
 - A wide genetic base allows plant breeders to develop new cultivars to meet challenges.
- First, we must identify sources of novel and diverse genetic material.
 - SNP markers obtained through Capture-Seq offer high coverage of the genome.

Methodology

1. Obtain 250 tissue samples – 221 *Prunus persica*, 29 other *Prunus*
2. Genotype using Capture-Seq method and call 134,424 SNPs.
3. Evaluate population structure.
 - a. STRUCTURE
 - b. PCA
 - c. AMOVA and UPGMA
4. Use SNP markers to design KASP primers for future studies.

Results

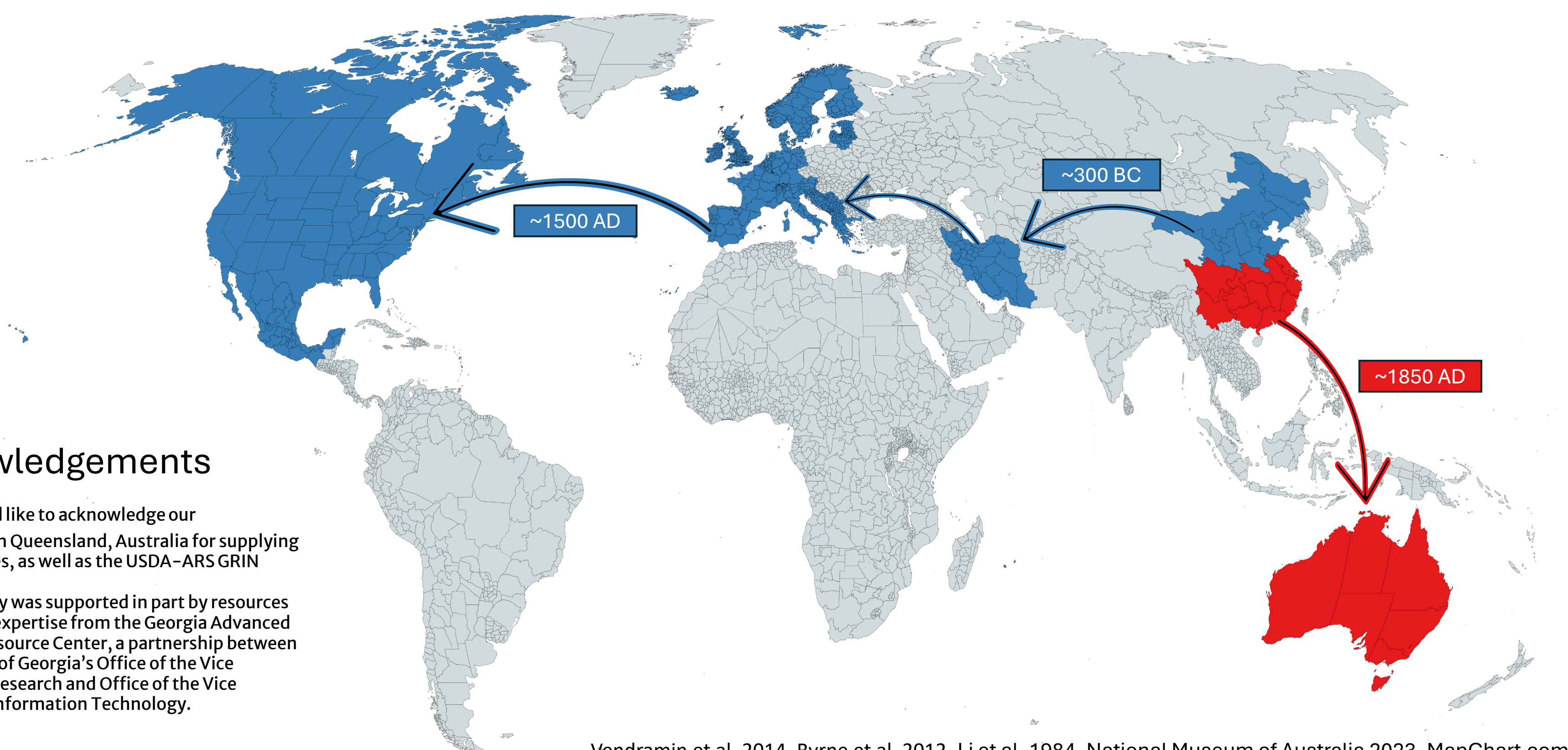
- SNP markers obtained via Capture-Seq can be used to differentiate between regional groups of peaches.
- European and North American populations are similar.
- Australian population is dissimilar to Euro/NA group.
- A small number of markers can also be used to differentiate between individuals and groups.
 - These can be converted into KASP markers



Acknowledgements

We would like to acknowledge our collaborators in Queensland, Australia for supplying us with samples, as well as the USDA-ARS GRIN repository.

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Vendramin et al. 2014, Byrne et al. 2012, Li et al. 1984, National Museum of Australia 2023, MapChart.com